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Sentiment Analysis of Visitor Reviews on Star Hotels in Manado City

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Abstract. Sentiment analysis is a technique of extracting the text data to analyze the opinions and evaluate to obtain the information. Sentiment analysis is performed by internet users on social media or online applications or websites to provide assessments or personal opinions. Tourism in North Sulawesi has grown by 600% in the past four years, and the rise of tourism has sent tourists flocking to the city of Manado. These travelers need a hotel that satisfies their desires, so they need to read about the hotel in the reviews on the hotel reservation service website. This takes a lot of time. To overcome existing problems, sentiment analysis applications were developed to make it easier for potential hotel users to find previous user responses. Additionally, data mining classification techniques are used to help hotel managers determine the satisfaction of previous hotel users using a Naive Bayes algorithm. There were 640 reviews data from January to March 2019 obtained from TripAdvisor website used in this study with the proportion of 70:30 for data training and testing respectively. The classification results are divided into five classes, namely excellent, good, average, poor, terrible. This classification is based on categories from the TripAdvisor website. The experimental result for five consecutive runs shows that Naïve Bayes obtained 76.20% accuracy, with an average of 70.55%. While the average precision is 70.57% and 99.85% for the recall.

Keywords: *Text Mining, Sentiment Analysis, Naïve Bayes, Hotels.*

1 Introduction

Tourism in North Sulawesi has grown by 600% in the past 4 years. This makes North Sulawesi appointed as the rising star of the year 2019 [1]. This can happen because it cannot be separated from the support of the local culture, festivals, culinary, tourist attractions, and existing infrastructure. The North Sulawesi Province has 4 cities and 11 regencies. Manado City is the capital city of North Sulawesi and is the city that has the highest number of hotels in this province, ranging from one to five-star hotels.

The hotel is one of the several facilities and infrastructure in the tourism sector. The increasing number of hotels in Manado City is due to the increasing number of tourists. Travelers are looking for a hotel that accommodates their needs and budget.

To find the information about the expected hotel, tourists must read the opinions or reviews of previous hotel users in the comments posted on the hotel booking service website. The information obtained is an opinion about the facilities, such as rooms, food, supporting facilities, and the services. Reading every comment there is, takes a lot of time. The comments given are personal thoughts or opinions that are influenced by emotion, this is often known as sentiment.

Bayesian classification is based on Bayes' theorem. When applied to large databases, Bayesian classification also shows high performance in terms of high accuracy and high convergence speed. In Naive Bayes Classifier, the attribute value in each class does not depend on other attribute values. Bayes's rule is used to classify new instances selecting the most likely ones that have been generated. In addition, Bayesian classification is also the simplest and most widely used classification method [2].

Sentiment towards hotels is high because hotel users express their opinions about the facilities and services on the hotel reservation service website, which is a place to get correct information according to the facts. As in [3], the author reviews the online ticket sales and hotel booking website, namely Agoda, to classify user comments into positive or negative classes using the Naive Bayes classifier. The purpose of sentiment analysis in this study is to obtain information from user reviews that are useful in improving the service quality of each existing hotel. Meanwhile, in this study [4] they implemented the naive Bayes algorithm to analyze sentiment on XYZ hotel user comments on the Agoda site, with the aim of helping XYZ hotel in finding the meaning of the comments as a whole.

In this paper, we build an application by the implementation of the Naive Bayes algorithm to make it easier for tourists and hotel managers to analyze the comments on the hotel. The contribution of this study is an application of sentiment analysis for four-star hotels in Manado City which can assist tourists in finding hotels that suit their needs and assist hotel managers in knowing the visitor responses. These responses are used as information to improve the performance and services at the hotel. In this study, the comment data will be used as training data that will be crawled from the www.tripadvisor.com portal, and we used the Naive Bayes algorithm to classify comments into five classes according to the comments category on the TripAdvisor website.

The remainder of this paper is structured as follows, discuss the related works in Section II, and Section III presents the method for presenting the formulation of sentiment analysis classification. In addition, Section IV reports the evaluation, which accommodates our result of this study. Finally, Section V summarizes this work and recommends some future work.

2 Related Works

Travel has already ingrained itself into everyday life. However, tourism is not just about travel. Tourism has a broad definition and includes not only travel but also visits to tourist destinations and sites, usage of transportation infrastructure, services,

lodging, dining, entertainment, and social interactions between visitors and locals [5]. Tourism and hotel are closely related. Hotels are included in the main tourism facilities, which means that their lives and lives depend a lot on the number of tourists who come. The tourism industry as a building, the hotel sector as a pillar. Hotels are frequently rated in order to categorize them according to quality. The original goal of hotel rating was to tell visitors about the standard amenities they can expecting, but it has now evolved into an emphasis on the overall hotel experience. Hotel ratings system in general range from One-Star to Five-Star represents the lowest to highest score rating.

Text mining is to extract the knowledge or information from the text and data, this type is unstructured. The basic stage of text mining is to convert text into a semi-structured dataset to obtain the patterns and train a model to recognize patterns in new and unseen text. After converting unstructured text into semi-structured data, then applying any analytical technique for classification, grouping, and prediction. Text data is found in daily life, and the data can be processed according to the research objectives, such as articles in online media, online applications, websites, and so on. The implementation of text mining, such as an information retrieval system [6] and analyzing the sentiments based on status or comments from the public or users.

Generally, sentiment analysis is referred to as opinion mining. A technique to extract the text data to analyze the opinions and evaluate to obtain the information that is sentiment analysis. This is an area of science that analyzes people's opinions, feelings, evaluations, judgments, attitudes, and emotions about products, services, organizations, individuals, problems, events, issues, and their attributes [7]. Sentiment analysis can also be done in situations where help is needed to understand people's thinking patterns or tendencies. Sentiment analysis can be applied in various areas, from food, health, tourism, and even the economy. As in this research [8], sentiment analysis was carried out at online shops on social media, namely The BerryBenka Facebook page uses the Naive Bayes algorithm and the aim is to identify trends in public perception of online stores. The customer comment data was crawled from the BerryBenka Facebook page related to services, namely orders, delivery, complaints, etc. The results showed that the implementation of the Naive Bayes algorithm this time reached 93.7% and the results were shown in the form of a bar chart.

Furthermore, this paper [9] discusses a review from a business perspective, this study purposed to obtain the opinions or feelings of consumers towards their products, especially in this case the HARRIS Hotel and Conventions Malang. The data obtained were analyzed using the K-Modes method for clustering with the Bag of Nouns feature and then LVG2 to be classified with the score representation feature. Data usage is divided into two, namely balanced data which has a total of 154 with 77 positive and negative classes as well as 77, and 277 unbalanced data with 200 positive and 77 negative classes, the two data are compared with the final result. This classification test is carried out using a confusion matrix and the results are that precision is 89.2%, recall is 89.13%, and f1-score is 89.12% of balanced data, while unbalanced data gets a precision value of 87.38%, 73.07% for recall, and 76.46% for f1-score. From the classification results, it can be seen that balanced data get better

results than unbalanced data.

In addition, this study [10] raises the issue of opinions and experiences from hotel use regarding facilities, services, and travel distances. Currently, there are many hotel service websites, including reviews of existing hotels, ranging from facilities, services, and even prices to stay. In this study, 300 reviews were used on the TripAdvisor website, which consisted of 150 reviews of positive opinions and 150 reviews of negative opinions. To perform mining and data processing, the RapidMiner application with version 5.3.015 is used by the implementation of the Support Vector Machine (SVM) algorithm by using Particle Swarm Optimization feature selection. The negative class attributes are worst, broken, and terrible, while the positive class is good, amazing, and delicious. The accuracy results obtained using the SVM algorithm are 91.33%.

Moreover, sentiment analysis is to see public opinion related to a figure, such as political figures and celebrities. In [11], a sentiment analysis application implemented through Twitter of the 2019 presidential candidate for the Republic of Indonesia to assist classify the class or level of the public sentiment using the Naive Bayes method. The results of this study showed that the Jokowi-Ma'ruf Amin pair had a positive sentiment polarity score of 45.45% and a negative sentiment score of 54.55%, while the Prabowo-Sandiaga pair had a positive sentiment score of 44.32% received and negative. 55.68%. The combined data used for each presidential candidate's training data was then tested and found to be 81% accurate. In addition, comparisons were made using the SVM and K-Nearest Neighbor methods, the highest accuracy value was obtained using the Naive Bayes method.

From the related works presented above, clearly shows that each method gives different results. Each classification algorithm may perform different depending on the existing dataset due to the characteristics of the data. The implementation of Naïve Bayes Algorithm to solve several sentiment classification problems shows its superiority. Therefore, this study employs Naïve Bayes Algorithm as a solution method to classify comments into five classes hotel satisfaction levels.

3 Classification Method

In this section, we will elaborate on the algorithms used to perform sentiment analysis. Naive Bayes is a machine learning algorithm for classification problems. The algorithm is a statistical classification that can be used to predict the probability of belonging to a class, with a strong assumption that all predictors are independent of each other. In other words, it is assumed that the existence of a feature in a class is independent of the existence of other features in the same class. It is used for text classification with high-dimensional training datasets, some examples of which are spam filtering, sentiment analysis, and news article classification.

There are two main models as usual used in the Naive Bayes classification. These two models purposed to obtain the posterior probability of a class according to the distribution of words in a document. The divergence between these two models is that the first model takes word frequency into account, while the other model does not

take into account the frequency of words. The two models are multi-variate Bernoulli and multinominal models. To perform the Naive Bayes classification, Equation 1 can be used which is a formula from Bayes' theorem, where X is the data with an unknown class and hypothesis for a specific class as denoted by H. Further, $P(H|X)$ is the probability of hypothesis H according to condition X, or the posterior probability, and the probability of hypothesis H is $P(H)$, or the prior probability. Moreover, $P(X|H)$ is the probability X according to the hypothesis H, and the probability X is represented by $P(X)$.

$$\boldsymbol{P(H|X) = \frac{P(H|X) * P(H)}{P(X)}} \quad (1)$$

$$\boldsymbol{P(C|F_1, \dots, F_n) = \frac{P(C) * P(F_1,\dots,F_n|C)}{P(F_1,\dots,F_n)}} \quad (2)$$

To explain the Naive Bayes, it is important to note that the classification process involves a series of instructions to determine which class is appropriate for the sample being analyzed. Therefore, the Naive Bayes is adjusted as shown in Equation 2, where C denotes the class and $F_1, F_2, \dots, F_n$ shows the properties of the instructions required to perform the classification. Thus, the equation explains that the probability of entering a sample with certain properties in class C (the posterior) is the probability that class C occurs (before the inclusion of the sample, often called prior), multiplied by the probability of occurrence of sample properties in class C which is called the likelihood, divided with the opportunity for the emergence of the universal sample properties or what is called evidence. Eventually, the formula can be written simply as follows:

$$\boldsymbol{Posterior = \frac{Prior * Likelihood}{Evidence}} \quad (3)$$

For example, we have training data of the hotel reviews or comments, as shown in Table 1. Further, the text preprocessing phase is carried out, namely tokenizing or lemmitazion, which is shown in Table 2. However, the sample text preprocessing table is not shown in its entirety.

Table 1. An Example of Training Data

Hotel Name	*Comments*	*Class*
Novotel	*We love the hotel room, it is clean and smell good.*	*Excellent*
Novotel	*Bad Hotel Ever. Bad Security.*	*Terrible*
Aryaduta	*Great location, but The Hotel need Renovation Here*	*Average*
Aryaduta	*Good hotel in nice location. The room was so nice and spacious. I love the breakfast.*	*Good*
Aryaduta	*Arya Duta is the best hotel in town, Good serviced, so clean and fresh.*	*Excellent*
Aryaduta	*The Hotel is one of old hotel in Manado. Yet, the room still nice.*	*Average*
Novotel	*Overall, this hotel is old, smelly, dirty floor in My Room.*	*Terrible*
Aryaduta	*Room and bathroom is spacious, however, not clean*	*Poor*
Aryaduta	*"Highly recommended hotel in Manado. The location, room, pool and service are Excellent!!"*	*Excellent*
Aryaduta	*"nice hotel good facilities staff are also very fiendly*	*Good*

Table 2. Tokenizing

Hotel Name	*Comments*	*Class*
Novotel	[*we, love, the, hotel, room, it, is, clean, and, smell, good*]	*Excellent*
Novotel	[*bad, hotel, ever, bad, security*]	*Terrible*
Aryaduta	[*great, location, but, the, hotel, need, renovation, here*]	*Average*
Aryaduta	[*good, hotel, in, nice, location, the, room, was, so, nice, and, spacious, i, love, the, breakfast*]	*Good*
Aryaduta	[*arya, duta, is, the, best, hotel, in, town, good, serviced, so, clean, and, fresh*]	*Excellent*
Aryaduta	[*the, hotel, is, one, of, old, hotel, in, manado, yet, the, room, still, nice*]	*Average*
Novotel	[*overall, this, hotel, is, old, smelly, dirty, floor, in, my, room*]	*Terrible*
Aryaduta	[*room, and, bathroom, is, spacious, however, not, clean*]	*Poor*
Aryaduta	[*highly, recommended, hotel, in, manado, the, location, room, pool, and, service, are, excellent*]	*Excellent*
Aryaduta	[*nice, hotel, good, facilities, staff, are, also, very, fiendly*]	*Good*
Aryaduta	[*ac, not, working, and, additional, dirty, old, wall, unit, blows, straight, on, bed*]	*Poor*
Novotel	[*this, hotel, is, old, dirty, floor, in, balcony*]	*Terrible*

Moreover, calculate the probability value of a class by dividing the number of class data by the total number of documents that exist, resulting that:

$$\boldsymbol{P(Terrible)} = \frac{3}{12} = 0{,}25$$

$$\boldsymbol{P(Poor)} = \frac{2}{12} = 0{,}166$$

$$\boldsymbol{P(Average)} = \frac{2}{12} = 0{,}166$$

$$\boldsymbol{P(Good)} = \frac{2}{12} = 0{,}166$$

$$\boldsymbol{P(Excellent)} = \frac{3}{12} = 0{,}25$$

Meanwhile, suppose we have the testing data as follows: "*Good Hotel*" for Hotel Aryaduta. Accordingly, we calculate the value of the test data for the existing class. For the terrible class are $P(Terrible|Good) = \frac{(0+6)*(0.5)}{(5+6)} = 0.272$, $P(Terrible|Hotel) = \frac{11.33}{19} = 0.596$. While for the poor class are $P(Poor|Good) = 0.272$, $P(Poor|Hotel) = 0.438$, and for the average class are $P(Average|Good) = 0.272$, $P(Average|Hotel) = 0.543$. $P(Good|Good) = 0.454$, $P(Good|Hotel) = 0.543$ for the good class, and $P(Excellent|Good) = 0.454$, $P(Excellent|Hotel) = 0.596$ for the excellent class.

Furthermore, the results of all calculations of the probability are multiplied by the probability value of every class in the training data.

$$\boldsymbol{P(Terrible)} = 0{,}272 * 0{,}596 = 0{,}162.$$

$$\boldsymbol{P(Poor)} = 0{,}272 * 0{,}438 = 0{,}119.$$

$$\boldsymbol{P(Average)} = 0{,}272 * 0{,}543 = 0{,}147$$

$\boldsymbol{P(Good)} = 0{,}454 * 0{,}543 = 0{,}246.$
$\boldsymbol{P(Excellent)} = 0{,}454 * 0{,}596 = \mathbf{0,270}.$
Based on the calculations that have been carried out, it is concluded that the existing testing data is classified in the excellent class.

In addition, we use the confusion matrix to measure the classification results produced by the algorithm. There are generally two types of confusion matrices. However, in this study, we used a multi-class (5x5) confusion matrix. A True Positive is the positive value of the result with the actual classification result and the result of the same classification class, while the negative value of the classification result with the correct class is denoted by the True Negative, and the negative values that are classified as positive are False Positive. However, the positive values are misclassified as negative is False Negatives.

Accuracy is the percentage of the proximity of the measured value or the value of the classification results and the actual value. Whereas, Precision is a measure of certainty, namely the percentage of classification results that are in a positive class, and a Recall is a measure of completeness, namely the percentage of positive values that have a positive class as well. The following is the formula for calculating accuracy, precision, and recall:

$$\boldsymbol{Accuracy} = \frac{\boldsymbol{Number\ of\ correct\ classification}}{\boldsymbol{Total\ number\ of\ classifications}} * \mathbf{100}\% \quad (4)$$

$$\boldsymbol{Precision} = \frac{\boldsymbol{True\ Positive}}{\boldsymbol{True\ Positive + False\ Positive}} \quad (5)$$

$$\boldsymbol{Recall} = \frac{\boldsymbol{True\ Positive}}{\boldsymbol{True\ Positive + False\ Negative}} \quad (6)$$

There are rules in the multiclass confusion matrix, namely *TP* is the value of the classification results which have the same class as the actual data. In precision calculations, *TP* and *FP* are true positive and false positive predictive values for the class being classified. *FP* is also the summation of the corresponding column values outside the *TP* value. While in the recall calculation, *TP* and *FN* are positive true and false negative predictive values for the class being classified, *TP+FN* is the total test cases from the class being classified. To determine positive and negative values can be seen according to columns and rows, positive values are values based on columns, and negative values are values based on rows. Furthermore, the precision value is taken based on the class column of the classification results and recall is based on the row of the classification results.

4 Performance Evaluation

In this section, we present the results and discussion of the applying sentiment analysis using the Naive Bayes algorithm. This application can support the prospective hotel users to find out the information about the desired hotel, and managers of hotels to find out the level of predecessor hotel user's satisfaction through

the results of the classification of comments or reviews. This application has four main features, namely crawl, text preprocessing, analyze and chart. The crawl feature is used to clean data from various punctuation marks and icons, convert all capital letters to lowercase and remove affixes for each word. Whereas, the analyze feature is for testing, here the testing is divided into two, namely single testing and multiple testing. Further, the chart feature is used to view the diagram of the accuracy results from the previous testing features.

In this testing, we used 640 reviews are crawling from the TripAdvisor website as training data, by January 2019 to March 2019, then the crawled data was stored in the database. Moreover, the weight of each word in the comment is calculated. Furthermore, data cleaning is carried out from the various punctuation marks, symbols, changing all capital letters to lowercase letters and removing affixes, as well as removing stopwords. After that, calculate the probability of each word in one comment, and apply the Naive Bayes algorithm on the previous data model. The data is divided based on the proportion of 70:30, for training data used 70% of the total data and 30% for testing data. After the testing, it will get the results of accuracy, precision, and recall as well as a chart to show the results of accuracy.

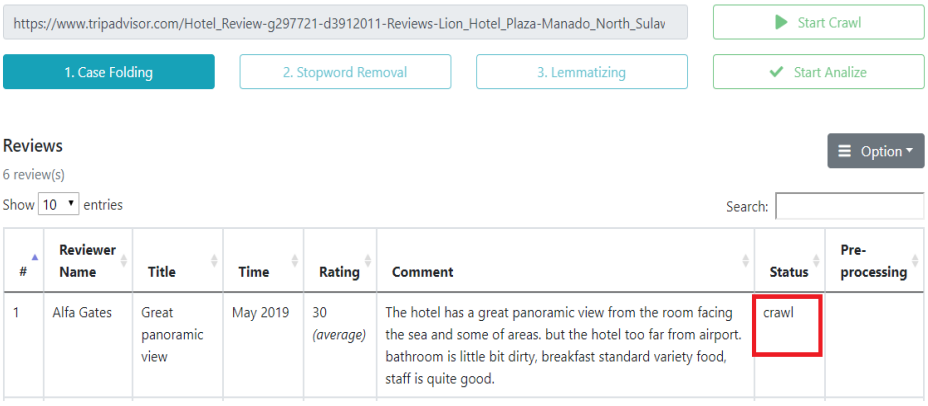


Figure 1. The Interface of Crawling the Reviews on TripAdvisor Website

In Figure 1 shows the appearance of the application in crawling reviews or comments from the TripAdvisor website, while the appearance of the application when doing text preprocessing, namely the case folding, stopwords, and lemmatization processes, as shown in Figure 2. Furthermore, Figure 3 shows the single test result which is the result of calculating the comments entered by the user on the single test page.

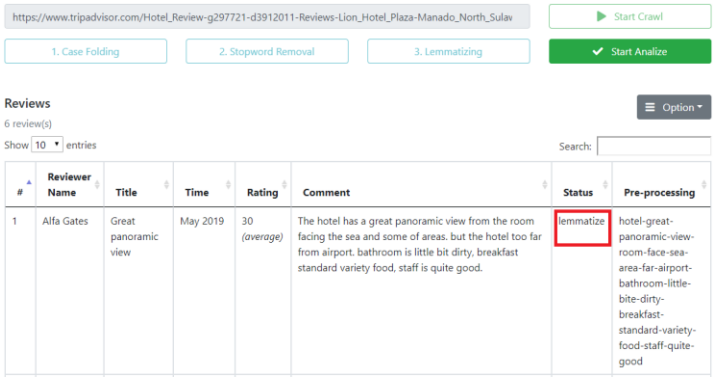


Figure 2. The Interface of Text Preprocessing

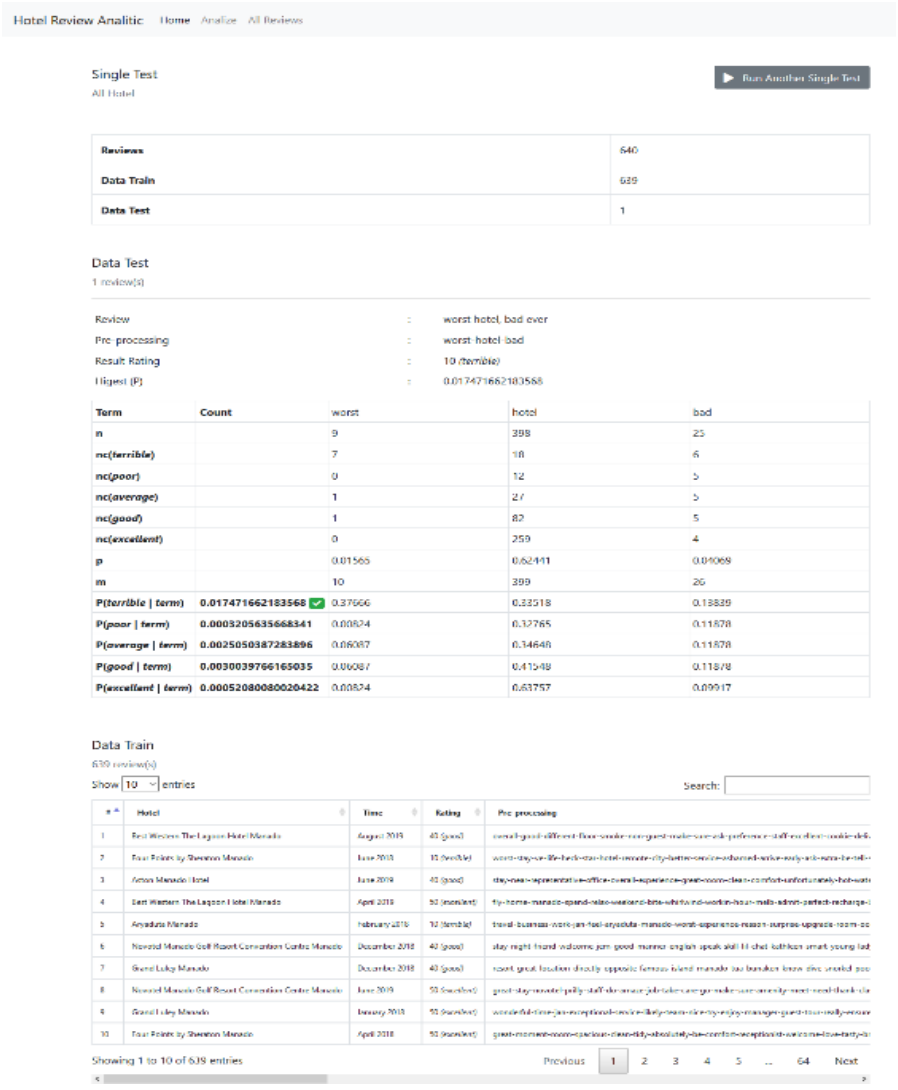


Figure 3. The Result of Single Test

Meanwhile, Figure 4 is the analysis result, which is from the multiple testing, the test results are taken as 30% of the training data, as well as provides the results of accuracy, precision, and recall. This analysis result displays a visualization in the form of a bar chart that shows the results of the classification, and a pie chart that displays the percentage of accurate results. Likewise, Figure 5 is the interface to show the overall reviews.

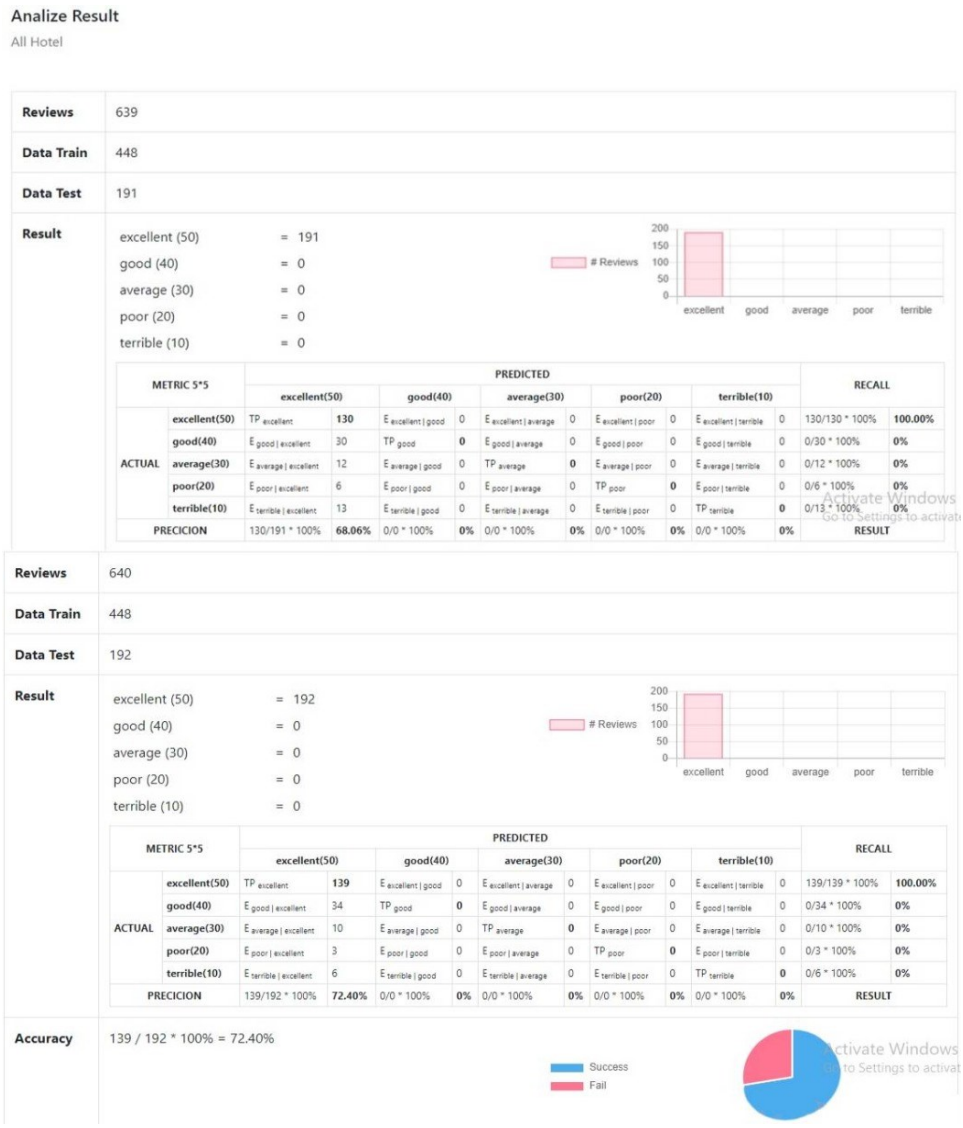


Figure 4. The Analyze Result of Multiple Testing

We used five times testing, the aim is to find the comparison of the testing results. Each testing carried out gets different results, due to the training data being taken randomly with the ratio of 70:30. The comparison table of the testing results

from the 1st test to the 5th test, is presented in Table 3.

Table 3. The Comparison of Testing Results

Testing	Accuracy	Precision	Recall
1	65,63%	65,97%	100%
2	70,83%	71,20%	100%
3	71,88%	72,63%	99,28%
4	68,23%	68,23%	100%
5	76,20%	74,80%	100%
Average	70,55%	70,57%	99,856%

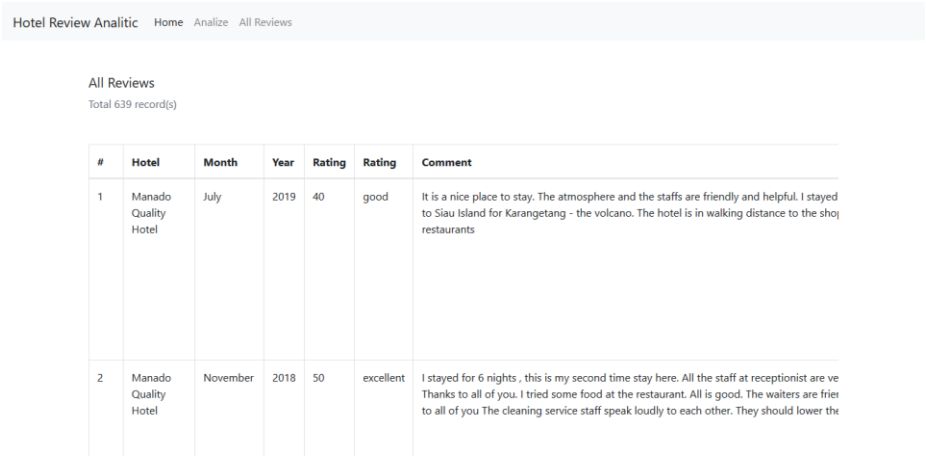


Figure 5. The Interface of the Overall Reviews

5 Conclusions

This paper proposes an application of sentiment analysis to four-star hotels in Manado City. Our application uses hotel user comments are taken by crawling data from the TripAdvisor website. We implement the Naive Bayes algorithm to analyze comments or reviews which consist of excellent, good, average, poor, and terrible classes. The classification process obtains an average accuracy result of 70.55% with the highest accuracy, i.e. 76.20%. This result shows that the model is good and reliable enough in correctly classified the reviews. While for the precision, the best is 72.63% with the average 70.57%, and the average of recall is 99.85% from 5 times of testing. The number of precision is not so vary from the accuracy which means that the quality of the model is good enough in predicting each category of the reviews. However, often there is an inverse relationship between precision and recall, where it is possible to increase one at the cost of reducing the other. The recall shows that out of all the times each review category should have been predicted, 99.85 % of the labels were correctly predicted. The greater number of recall is preferred with the trade off with accuracy and precision results so that we may be able to know all relevant result for each of the category review correctly classified by Naive Bayes.

We can conclude that our model is performing well. The results for 5 consecutive tests show consistency with very small variance. The best execution time is on Google Chrome version 79.0.3945.130, i.e. less than 2 seconds. For further study, we suggest that the crawling review can be done simultaneously for all hotels and for the text pre-processing process can be concurrently maximized so it doesn't take a long time. Comparison with several baseline classifiers is worth to do.

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